6.01	Algebraic expressions			
6.01a	Algebraic terminology and proofs	Understand and use the concepts and vocabulary of expressions, equations, formulae, inequalities, terms and factors.	Recognise the difference between an equation and an identity, and show algebraic expressions are equivalent. e.g. show that $(x+1)^2 + 2 = x^2 + 2x + 3$ Use algebra to construct arguments.	Use algebra to construct proofs and arguments. e.g. prove that the sum of three consecutive integers is a multiple of 3.
6.01b	Collecting like terms in sums and differences of terms	Simplify algebraic expressions by collecting like terms.		

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to…	Foundation tier learners should also be able to…	Higher tier learners should additionally be able to…
6.01c	Simplifying products and quotients	Simplify algebraic products and quotients. e.g. $a \times a \times a = a^3$ $2a \times 3b = 6ab$ $a^2 \times a^3 = a^5$ $3a^3 \div a = 3a^2$ *[see also Laws of indices, 3.01c]*		Simplify algebraic products and quotients using the law of indices. e.g. $a^{\frac{1}{2}} \times 2a^{-3} = 2a^{-\frac{5}{2}}$ $2a^2b^3 \div 4a^{-3}b = \frac{1}{2}a^5b$
6.01d	Multiplying out brackets	Simplify algebraic expressions by multiplying a single term over a bracket. e.g. $2(a+3b) = 2a+6b$ $2(a+3b)+3(a-2b) = 5a$	Expand products of two binomials. e.g. $(x-1)(x-2) = x^2-3x+2$ $(a+2b)(a-b) = a^2+ab-2b^2$	Expand products of more th two binomials. e.g. $(x+1)(x-1)(2x+1)$ $= 2x^3+x^2-2x$
6.01e	Factorising	Take out common factors. e.g. $3a-9b = 3(a-3b)$ $2x+3x^2 = x(2+3x)$	Factorise quadratic expressions of the form x^2+bx+c. e.g. $x^2-x-6 = (x-3)(x+2)$ $x^2-16 = (x-4)(x+4)$ $x^2-3 = (x-\sqrt{3})(x+\sqrt{3})$	Factorise quadratic expressi of the form ax^2+bx+c (wh $a \neq 0$ or 1) e.g. $2x^2+3x-2 = (2x-1)(x+$
6.01f	Completing the square			Complete the square on a quadratic expression. e.g. $x^2+4x-6 = (x+2)^2-10$ $2x^2+5x+1 = 2\left(x+\frac{5}{4}\right)^2 -$

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6.01g	Algebraic fractions			Simplify and manipulate algebraic fractions. e.g. Write $\frac{1}{n-1}+\frac{n}{n+1}$ as a single fraction. Simplify $\frac{n^2+2n}{n^2+n-2}$.	A1, A4